MILITARY SPECIFICATION

FURNACE; ELECTRIC, BOX TYPE, UNIVERSAL

This specification has been approved by
the Bureau of Aeronautics, Department of the Navy

1. SCOPE AND CLASSIFICATION

1.1 SCOPE: This specification is for a box type electrical furnace designed for use in the heat treatment of aluminum alloy or alloy steel by means of either radiation or convection in which a working temperature from 250°F. to 2000°F. is required.

2. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATIONS

2.1 GENERAL: The following specifications and publications of the issue in effect on the date of invitation for bids, form a part of this specification.

2.2 SPECIFICATIONS:

Navy Department Specifications

General Specifications for Inspection of Material.

2.3 AVAILABILITY OF SPECIFICATIONS AND PUBLICATIONS

2.3.1 When requesting specifications and publications refer to both title and number.

2.3.2 Copies of this specification and applicable specifications may be obtained upon application to the Commander, U. S. Naval Air Development Center, Johnsville, Pennsylvania, Attention Technical Records Department.

3. REQUIREMENTS: The furnace shall be electrically operated and consist of the furnace body, hearth plate, radiant heating element, convection unit, insulation and control equipment as described herein.

3.1 Type, Size and Construction. -

3.1.1 Furnace Body. - Furnace shall be of rigid and substantial construction and shall be built in accordance with best modern commercial furnace practice. The very best grade of refractory and insulating materials shall be used throughout. Workmanship and materials shall be first class in every respect and the furnace shall be capable of economical operation and ease of handling. Two baskets, each partitioned for heat treating 4 different size rivets shall be provided. The furnace shall be designed and constructed to operate at a maximum temperature of 2000°F. and at any working temperature ranging from 250°F. to 2000°F. The temperature variation throughout the furnace hearth and charge shall not exceed plus or minus of 8°F. for any change within the capacity of the furnace and within the useful working temperatures of the furnace, under saturated conditions. The heating area shall be 18" wide, 36" deep and 18" high

3.1.2 The outer shell of the furnace shall be made of steel plate approximately 1/8" in thickness, properly reinforced and welded. The furnace shall be supported on a structural steel framework at a suitable height above the floor level to permit easy charging and discharging.

3.1.3 The front of the furnace shall be made up of a cast iron or steel front plate properly ribbed to prevent warpage. A cast iron or steel heat-insulated door, properly counter-weighted and provided with a lever for easy opening, shall be provided. The door shall be provided with jambs for sealing when in the closed position. It shall overlap the brick facing of the vestibule on all four sides for effective heat insulation.

3.2 Hearth Plate. - A cast heat-resisting nickel-chrome hearth plate shall be provided having upturned flanges 2" high on sides and back. It shall be sufficiently heavy to sustain any load that may be placed on it under the maximum temperature conditions, without distortion. It shall be made of an alloy containing not less than 35% nickel, 15% chrome and the balance iron.

3.3 Radiant Heating and Convection Units: Both heating elements shall operate on a 220 volt, 3 phase, 60 cycle power supply as a balanced three phase load. The capacity of the heating elements and the thickness of the refractory and insulating walls shall be such as to permit heating the empty furnace from room temperature to 1500°F. in one and one-half hours. It shall recover this temperature after an

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overnight shut down of 16 hours in 45 minutes.

3.3.1 Radiant Heating Elements: The radiant heating elements of the furnace shall be mounted on the walls, arch, door and under the hearth plate, or any combination which will provide a uniform temperature throughout the hearth, and shall be easily removable and be so distributed that heating will be uniform through the furnace chamber. All elements shall be of drawn nickel-chromium wire or ribbon of an alloy containing approximately 80% nickel, 20% chromium and less than 1% iron as an impurity. The temperature difference between the heating element surface at the bend and the center of the furnace under saturated conditions at 2000°F. shall not exceed 100°F. Elements shall be secured in a manner to preclude the possibility of becoming loose or working out of place under all operating conditions. The terminals of the elements shall have at least twice the cross sectional area of the heating elements to prevent over-heating where they must pass through the insulated walls and they shall be provided with solderless connectors to power leads. The radiant heating unit shall have an input of approximately 33 KW and shall operate the furnace continuously at any working temperature from 1250°F to 2000°F. The radiation loss of the furnace shall not exceed 7 KWH per hour at a saturated temperature of 1550°F.

3.3.2 Convection Unit. - For the heat treatment of aluminum alloys, where radiant type heating elements are not permitted, a convection heating unit of the recirculating type shall be supplied. This unit shall be constructed as an integral part of the furnace structure and shall consist of an insulated chamber containing suitable heating elements, a horizontal pressure type fan having two air-cooled bearings and a motor and base for driving the fan. Suitable electrical and mechanical devices shall be provided so that the power cannot be applied to the convection chamber heating elements except when the fan is in operation. Nickel-chrome dampers shall be provided, so that the convection unit may be sealed from the heating chamber when the furnace is operated in excess of 1250°F. All parts of the convection heating unit must be designed to operate at a maximum temperature for at least 1250°F. The heating elements of the convection heating unit shall be mounted on a nickel-chrome frame, proper insulation and so arranged that the entire assembly may be removed from the chamber without dismantling the convection unit.

3.4 Insulation. - Furnace shall be effectively insulated by high, medium and low temperature materials of ample thickness to insure a minimum heat loss through radiation. All insulation shall be in the form of preformed slabs or bricks. Powdered insulation is not acceptable.

3.5 Control Equipment:

3.5.1 Indicating Controlling Pyrometer. - The control of the furnace temperature shall be automatic. This control shall be accomplished by means of an indicating controlling pyrometer of the millivolt type employing an electronic circuit between the temperature indicating element and the control mechanism of the instrument. This instrument shall be equipped with cold junction compensation and thermocouple bead protection. It shall be enclosed in a dust proof case and provided with a temperature setting device. It shall be calibrated for chromel-alumel thermocouple of and shall be furnished with #8 gauge chromel-alumel thermocouple of proper length for the furnace provided with suitable head and head casting, 25 feet of lead wire shall be provided to connect thermocouple to pyrometer.

3.5.2 Magnetic Contactors. - Two magnetic contactors together with suitable relays shall be provided. One contactor will control the radiant heating elements and the other contactor will control the convection heating elements. They shall be so connected that only one of the contactors may be closed at one time. Suitable pushbutton stations for the control circuits, shall be provided.

3.5.3 Mounting Panel. - A steel instrument mounting panel shall be provided, completely wired, and arranged for flush mounting of the control pyrometer and the excess temperature cutout. The panel shall also be equipped with a suitable magnetic starter for the motor of the convection unit. The pushbutton stations for the control circuit shall be mounted and wired on the panel.

3.5.4 Excess Temperature Cut-Out. - An excess temperature cut-out shall be provided that may be set at any predetermined temperature in excess of the operating temperature, to protect the furnace against burnout in case of failure of a control pyrometer. This cut-out shall be actuated by a separate chromel-alumel thermocouple located in the heating chamber. The thermocouple shall be supplied with 25 feet of lead wire.

3.6 Marking. - A brass nameplate shall be mounted on the furnace. It shall be stamped, or deeply engraved to show the manufacturer's name, serial number, identification symbols, voltage and kilowatt rating, and date of manufacture A blank space shall be provided for stamping the name of the station on which installed and the requisition number and date.

4. INSPECTION AND TEST

The following inspection shall be made at the point of manufacture upon the furnace and its control equipment. Each furnace shall be subjected to a thorough examination by the Government Inspector in accordance with General Specifications for Inspection of Material to ascertain that the materials, workmanship, design, etc., are in conformity with the specifications.

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The following inspection shall be made at the manufacturer's plant. Each furnace and control shall be subjected to an operating test in the presence of the Government Inspector to determine that the radiation losses, time to reach specified temperatures, temperature gradient between resistor and furnace and uniformity of temperature throughout the furnace chamber are in accordance with the specifications. The temperature uniformity shall be determined by inserting a calibrated thermocouple into the chamber, with furnace couple in its place, and making explorations for temperature gradient.

5. Packing. - Each unit shall be securely packed complete in one crate in a manner to insure acceptance by common carriers in accordance with the current issue of the Consolidated Freight Classification Rules.

6. NOTES

6.1 INTENDED USE. - These furnaces are to be used in the heat treatment of aluminum alloy or alloy steel requiring a working temperature range from 250°F to 2000°F.

6.2 MISCELLANEOUS NOTES. - This specification covers stock numbers V40-F-1200 furnaces electrical box type universal appearing in the Department of the Navy, Bureau of Aeronautics Catalog of Navy Material, Aviation Supply Office section.

6.3 Failure due to defective design, material or workmanship, which may develop prior to the completion of the first year of actual service shall be either repaired or replaced at the expense of the contractor.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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